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**Title** – E-Commerce Sales and Customer Behavior Analysis

**Project Domain** – E-Commerce

**Dataset Timeline** – 2023 - 2025

**Submission Date** – 01/05/2026

**Mentor Name** – Kumaran M.

**Raw Dataset Link** –  [E-commerce\\_sales\\_dataset.csv](#)

**Cleaned Dataset Link** –  [E-commerce\\_sales\\_dataset\\_Cleaned.xlsx](#)

# E-Commerce Sales and Customer Behavior Analysis

## Objective:

1. To identify meaningful patterns, trends, and insights in e-commerce sales transactions using exploratory data analysis (EDA)
2. To clean, transform, and preprocess the dataset by handling missing values and preparing the data for accurate analysis
3. To analyze customer purchasing behavior, sales performance, and regional distribution of transactions
4. To create clear and informative visualizations using Python libraries such as Matplotlib and Seaborn
5. To derive actionable insights and meaningful conclusions that can support data-driven decision making in an e-commerce environment

## Outcome:

The project uncovers key patterns and trends in e-commerce transactions through data cleaning, exploratory analysis, and visualization. The analysis provides meaningful insights into sales performance and customer behavior, supporting informed and data-driven decision making.

## Dataset Information:

**Source:** Kaggle

**Timeline:** Data collected during 2023 to 2025

## Dataset Description:

The dataset contains e-commerce transaction records including product details, customer information, payment methods, order dates, and sales metrics. It consists of 34,500 records and 17 features, providing comprehensive data for analyzing sales trends, customer behavior, and regional performance.

**order\_id** – It identifies the unique ID of each order placed in the e-commerce platform.

**customer\_id** – It represents the unique identifier assigned to each customer.

**product\_id** – It indicates the unique ID of the product purchased in the transaction.

**category** – It shows the category to which the purchased product belongs.

# E-Commerce Sales and Customer Behavior Analysis

price – It represents the price of a single unit of the product.

discount – It indicates the discount applied to the product during the purchase.

quantity – It shows the number of units of the product purchased in the order.

payment\_method – It tells us the method used by the customer to complete the payment.

order\_date – It represents the date on which the order was placed.

delivery\_time\_days – It indicates the number of days taken to deliver the order.

region – It shows the geographical region where the order was delivered.

returned – It indicates whether the product was returned by the customer or not.

total\_amount – It represents the total amount paid by the customer for the order.

shipping\_cost – It shows the cost incurred for delivering the product to the customer.

profit\_margin – It indicates the profit earned from the sale of the product.

customer\_age – It represents the age of the customer who made the purchase.

customer\_gender – It shows the gender of the customer who placed the order

## Type of Analysis

### Descriptive Analysis

Summarizing the dataset to understand basic information such as sales distribution, customer demographics, product categories, and transaction patterns.

### Diagnostic Analysis

Analyzing relationships in the data to identify reasons behind trends such as variations in sales across regions, the impact of discounts on revenue, or differences in customer purchasing behavior.

### Prescriptive Analysis

Providing data-driven recommendations based on the insights obtained, such as improving sales strategies, optimizing discount usage, or identifying high-performing product categories.

# E-Commerce Sales and Customer Behavior Analysis

## Stage 1 – Problem Definition and Dataset Selection

### 1. Problem Definition:

The rapid growth of e-commerce platforms has generated large volumes of transactional data. Businesses need to analyze this data to understand customer purchasing behavior, sales performance, and operational efficiency. However, identifying meaningful insights from raw transaction data can be challenging without proper data analysis techniques.

This project aims to analyze an e-commerce sales dataset to uncover patterns, trends, and relationships between various factors such as product categories, pricing, discounts, payment methods, and customer demographics.

### 2. Expected Outcome:

The expected outcome of this project is to extract meaningful insights from the dataset that help understand sales trends, customer behavior, and regional performance. The analysis will support data-driven decision-making and highlight potential areas for business improvement.

### 3. Dataset Selection:

#### Dataset Source

The dataset used in this project is the E-Commerce Sales Transactions Dataset, obtained from Kaggle, which contains simulated but realistic online retail transaction records.

#### Dataset Size

Rows: 34,500

Columns: 17

#### Dataset Features

The dataset contains information related to:

1. Orders and products
2. Customer demographics
3. Payment methods
4. Delivery information
5. Sales and profit metrics

These features enable comprehensive analysis of sales performance and customer purchasing patterns.

# E-Commerce Sales and Customer Behavior Analysis

## Python Implementation for E-Commerce Data Analysis

```
# importing libraries
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Loading dataset
```

```
df =
```

```
pd.read_csv('https://raw.githubusercontent.com/Vignesh-106/mydataset/refs/heads/main/E-commerce_sales_dataset.csv')
```

|       | order_id | customer_id | product_id | category    | price  | discount | quantity | payment_method | order_date | delivery_time_days | region | returned | total_amount | shipping_cost |
|-------|----------|-------------|------------|-------------|--------|----------|----------|----------------|------------|--------------------|--------|----------|--------------|---------------|
| 0     | O100000  | C17270      | P234890    | Home        | 164.08 | 0.15     | 1        | Credit Card    | 2023-12-23 | 4                  | West   | No       | 139.47       | 7.88          |
| 1     | O100001  | C17603      | P228204    | Grocery     | 24.73  | 0.00     | 1        | Credit Card    | 2025-04-03 | 6                  | South  | No       | 24.73        | 4.60          |
| 2     | O100002  | C10860      | P213892    | Electronics | 175.58 | 0.05     | 1        | Credit Card    | 2024-10-08 | 4                  | North  | No       | 166.80       | 6.58          |
| 3     | O100003  | C15390      | P208689    | Electronics | 63.67  | 0.00     | 1        | UPI            | 2024-09-14 | 6                  | South  | No       | 63.67        | 5.50          |
| 4     | O100004  | C15226      | P228063    | Home        | 16.33  | 0.15     | 1        | COD            | 2024-12-21 | 6                  | East   | No       | 13.88        | 2.74          |
| ...   | ...      | ...         | ...        | ...         | ...    | ...      | ...      | ...            | ...        | ...                | ...    | ...      | ...          | ...           |
| 34495 | O134495  | C13553      | P231412    | Fashion     | 10.44  | 0.20     | 1        | Debit Card     | 2025-08-13 | 3                  | North  | No       | 8.35         | 3.04          |
| 34496 | O134496  | C11349      | P218266    | Beauty      | 4.79   | 0.10     | 5        | Debit Card     | 2023-12-04 | 6                  | South  | Yes      | 21.56        | NaN           |
| 34497 | O134497  | C12986      | P204345    | Electronics | 323.64 | 0.00     | 5        | Credit Card    | 2025-02-01 | 6                  | East   | No       | 1618.20      | 12.18         |
| 34498 | O134498  | C17196      | P217644    | Home        | 23.09  | 0.05     | 1        | Credit Card    | 2024-03-27 | 6                  | East   | No       | 21.94        | 5.61          |
| 34499 | O134499  | C13330      | P207047    | Home        | 76.84  | 0.00     | 1        | Debit Card     | 2025-09-11 | 6                  | East   | No       | 76.84        | 7.57          |

34500 rows x 17 columns

```
# Initial EDA
```

```
#First 5 rows
```

```
df.head()
```

|   | order_id | customer_id | product_id | category    | price  | discount | quantity | payment_method | order_date | delivery_time_days | region | returned | total_amount | shipping_cost |
|---|----------|-------------|------------|-------------|--------|----------|----------|----------------|------------|--------------------|--------|----------|--------------|---------------|
| 0 | O100000  | C17270      | P234890    | Home        | 164.08 | 0.15     | 1        | Credit Card    | 2023-12-23 | 4                  | West   | No       | 139.47       | 7.88          |
| 1 | O100001  | C17603      | P228204    | Grocery     | 24.73  | 0.00     | 1        | Credit Card    | 2025-04-03 | 6                  | South  | No       | 24.73        | 4.60          |
| 2 | O100002  | C10860      | P213892    | Electronics | 175.58 | 0.05     | 1        | Credit Card    | 2024-10-08 | 4                  | North  | No       | 166.80       | 6.58          |
| 3 | O100003  | C15390      | P208689    | Electronics | 63.67  | 0.00     | 1        | UPI            | 2024-09-14 | 6                  | South  | No       | 63.67        | 5.50          |
| 4 | O100004  | C15226      | P228063    | Home        | 16.33  | 0.15     | 1        | COD            | 2024-12-21 | 6                  | East   | No       | 13.88        | 2.74          |

```
#Last 5 rows
```

```
df.tail()
```

|       | order_id | customer_id | product_id | category    | price  | discount | quantity | payment_method | order_date | delivery_time_days | region | returned | total_amount | shipping_cost |
|-------|----------|-------------|------------|-------------|--------|----------|----------|----------------|------------|--------------------|--------|----------|--------------|---------------|
| 34495 | O134495  | C13553      | P231412    | Fashion     | 10.44  | 0.20     | 1        | Debit Card     | 2025-08-13 | 3                  | North  | No       | 8.35         | 3.04          |
| 34496 | O134496  | C11349      | P218266    | Beauty      | 4.79   | 0.10     | 5        | Debit Card     | 2023-12-04 | 6                  | South  | Yes      | 21.56        | NaN           |
| 34497 | O134497  | C12986      | P204345    | Electronics | 323.64 | 0.00     | 5        | Credit Card    | 2025-02-01 | 6                  | East   | No       | 1618.20      | 12.18         |
| 34498 | O134498  | C17196      | P217644    | Home        | 23.09  | 0.05     | 1        | Credit Card    | 2024-03-27 | 6                  | East   | No       | 21.94        | 5.61          |
| 34499 | O134499  | C13330      | P207047    | Home        | 76.84  | 0.00     | 1        | Debit Card     | 2025-09-11 | 6                  | East   | No       | 76.84        | 7.57          |

```
# shape of the dataset
```

```
df.shape
```

```
(34500, 17)
```

# E-Commerce Sales and Customer Behavior Analysis

```
#size of the dataset  
df.size
```

```
586500
```

```
#Column names  
df.columns
```

```
Index(['order_id', 'customer_id', 'product_id', 'category', 'price',  
      'discount', 'quantity', 'payment_method', 'order_date',  
      'delivery_time_days', 'region', 'returned', 'total_amount',  
      'shipping_cost', 'profit_margin', 'customer_age', 'customer_gender'],  
      dtype='object')
```

```
#info about Dataset
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 34500 entries, 0 to 34499  
Data columns (total 17 columns):  
#   Column                Non-Null Count  Dtype  
---  -  
0   order_id              34500 non-null  object  
1   customer_id           34500 non-null  object  
2   product_id            34500 non-null  object  
3   category               34500 non-null  object  
4   price                 34500 non-null  float64  
5   discount              32775 non-null  float64  
6   quantity              34500 non-null  int64  
7   payment_method        33810 non-null  object  
8   order_date            34500 non-null  object  
9   delivery_time_days    34500 non-null  int64  
10  region                33465 non-null  object  
11  returned              34500 non-null  object  
12  total_amount          34500 non-null  float64  
13  shipping_cost         32775 non-null  float64  
14  profit_margin         34500 non-null  float64  
15  customer_age          34500 non-null  int64  
16  customer_gender       33465 non-null  object  
dtypes: float64(5), int64(3), object(9)  
memory usage: 4.5+ MB
```

# E-Commerce Sales and Customer Behavior Analysis

#Descriptive stats of dataset

df.describe()

|       | price        | discount     | quantity     | delivery_time_days | total_amount | shipping_cost | profit_margin | customer_age |
|-------|--------------|--------------|--------------|--------------------|--------------|---------------|---------------|--------------|
| count | 34500.000000 | 32775.000000 | 34500.000000 | 34500.000000       | 34500.000000 | 32775.000000  | 34500.000000  | 34500.000000 |
| mean  | 119.391632   | 0.049230     | 1.490725     | 4.814203           | 170.008494   | 6.155670      | 28.116505     | 43.474377    |
| std   | 195.620477   | 0.069867     | 0.932270     | 1.242141           | 357.503014   | 2.387721      | 53.352947     | 14.980682    |
| min   | 1.010000     | 0.000000     | 1.000000     | 3.000000           | 0.820000     | 0.000000      | -6.200000     | 18.000000    |
| 25%   | 16.690000    | 0.000000     | 1.000000     | 4.000000           | 19.710000    | 4.430000      | 1.500000      | 31.000000    |
| 50%   | 45.660000    | 0.000000     | 1.000000     | 5.000000           | 56.820000    | 6.090000      | 10.550000     | 43.000000    |
| 75%   | 130.950000   | 0.100000     | 2.000000     | 6.000000           | 168.530000   | 7.830000      | 33.132500     | 56.000000    |
| max   | 2930.470000  | 0.300000     | 5.000000     | 13.000000          | 12931.800000 | 15.650000     | 1536.170000   | 69.000000    |

#Checking for missing values in the dataset

df.isnull().sum()

|                    |      |
|--------------------|------|
| order_id           | 0    |
| customer_id        | 0    |
| product_id         | 0    |
| category           | 0    |
| price              | 0    |
| discount           | 1725 |
| quantity           | 0    |
| payment_method     | 690  |
| order_date         | 0    |
| delivery_time_days | 0    |
| region             | 1035 |
| returned           | 0    |
| total_amount       | 0    |
| shipping_cost      | 1725 |
| profit_margin      | 0    |
| customer_age       | 0    |
| customer_gender    | 1035 |
| dtype: int64       |      |

# E-Commerce Sales and Customer Behavior Analysis

```
#Checking for duplicated values in the dataset  
df.duplicated().sum()
```

```
np.int64(0)
```

## Stage 2 – Data Cleaning and Pre-processing

```
# Handling missing values
```

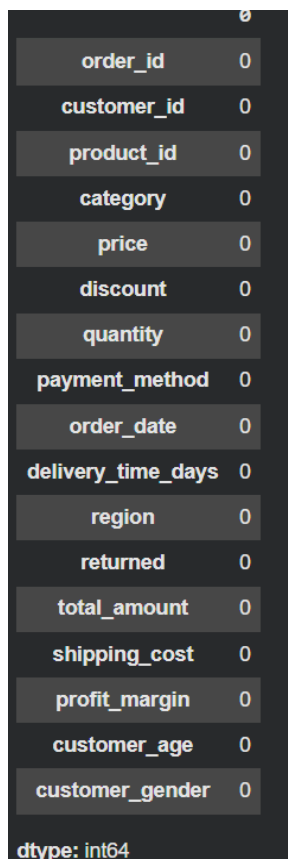
```
# Handling Numerical Missing Values - Discount and Shipping cost - Using 'Mean' as an  
imputation values since it provides an summarized value
```

```
df['discount'].fillna(df['discount'].mean(), inplace=True)  
df['shipping_cost'].fillna(df['shipping_cost'].mean(), inplace=True)
```

```
#Handling Categorical Missing Values - Payment method, Region and Gender - Using  
'Mode' for imputation beacuse the value obtained is most frequent
```

```
df['payment_method'].fillna(df['payment_method'].mode()[0], inplace=True)  
df['region'].fillna(df['region'].mode()[0], inplace=True)  
df['customer_gender'].fillna(df['customer_gender'].mode()[0], inplace=True)
```

```
df.isnull().sum()
```



|                    |   |
|--------------------|---|
| order_id           | 0 |
| customer_id        | 0 |
| product_id         | 0 |
| category           | 0 |
| price              | 0 |
| discount           | 0 |
| quantity           | 0 |
| payment_method     | 0 |
| order_date         | 0 |
| delivery_time_days | 0 |
| region             | 0 |
| returned           | 0 |
| total_amount       | 0 |
| shipping_cost      | 0 |
| profit_margin      | 0 |
| customer_age       | 0 |
| customer_gender    | 0 |

dtype: int64



# E-Commerce Sales and Customer Behavior Analysis

**Note - My dataset is free of duplicate values, ensuring the accuracy and reliability of the data for our analysis.**

```
#Treating Outliers  
df.describe().round(2)
```

|       | price    | discount | quantity | delivery_time_days | total_amount | shipping_cost | profit_margin | customer_age |
|-------|----------|----------|----------|--------------------|--------------|---------------|---------------|--------------|
| count | 34500.00 | 34500.00 | 34500.00 | 34500.00           | 34500.00     | 34500.00      | 34500.00      | 34500.00     |
| mean  | 119.39   | 0.05     | 1.49     | 4.81               | 170.01       | 6.16          | 28.12         | 43.47        |
| std   | 195.62   | 0.07     | 0.93     | 1.24               | 357.50       | 2.33          | 53.35         | 14.98        |
| min   | 1.01     | 0.00     | 1.00     | 3.00               | 0.82         | 0.00          | -6.20         | 18.00        |
| 25%   | 16.69    | 0.00     | 1.00     | 4.00               | 19.71        | 4.53          | 1.50          | 31.00        |
| 50%   | 45.66    | 0.00     | 1.00     | 5.00               | 56.82        | 6.16          | 10.55         | 43.00        |
| 75%   | 130.95   | 0.10     | 2.00     | 6.00               | 168.53       | 7.73          | 33.13         | 56.00        |
| max   | 2930.47  | 0.30     | 5.00     | 13.00              | 12931.80     | 15.65         | 1536.17       | 69.00        |

```
numerical_columns = df.select_dtypes(include=['int64', 'float64']).columns #getting only  
numerical columns for finding outliers
```

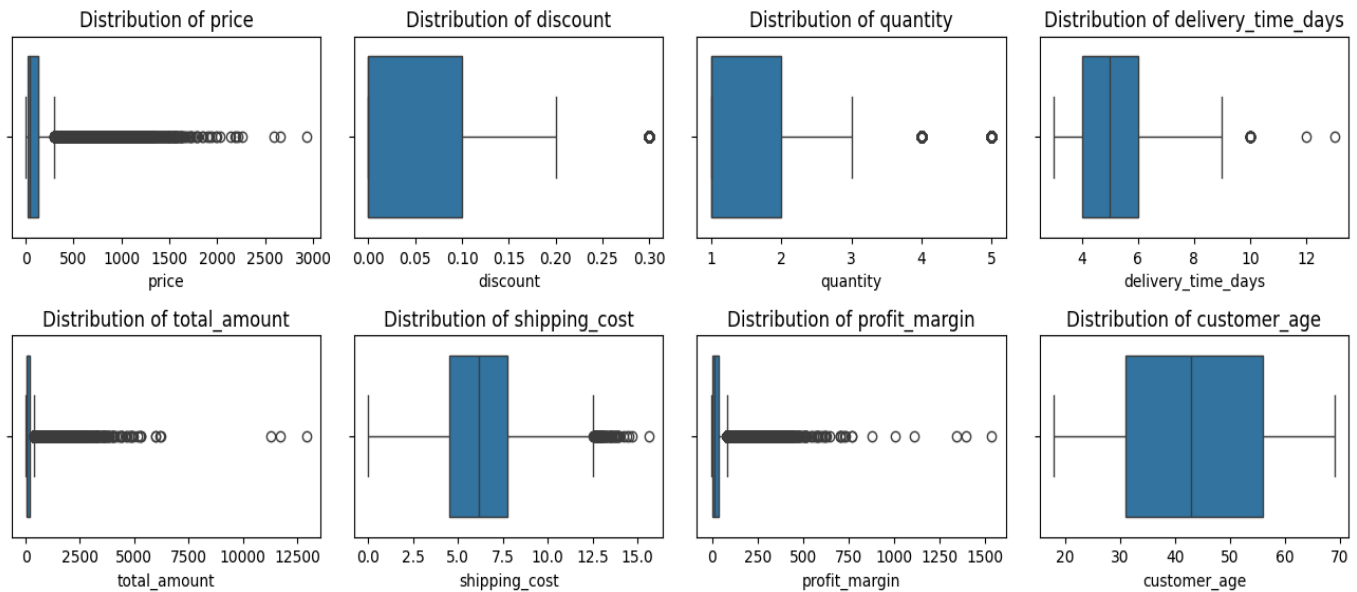
```
#identifying outliers using Boxplots  
plt.figure(figsize=(13,10))
```

```
for i, col in enumerate(numerical_columns, 1):  
    plt.subplot(4, 4, i) # 4 rows, 4 columns layout  
    sns.boxplot(x=df[col])  
    plt.title(f'Distribution of {col}')
```

```
plt.suptitle('Boxplots of Numerical Columns', fontsize=16, fontweight='bold', y=1.02)  
plt.tight_layout()  
plt.show()
```

# E-Commerce Sales and Customer Behavior Analysis

## Boxplots of Numerical Columns



#identifying outliers using KDEplots

plt.figure(figsize=(13,10))

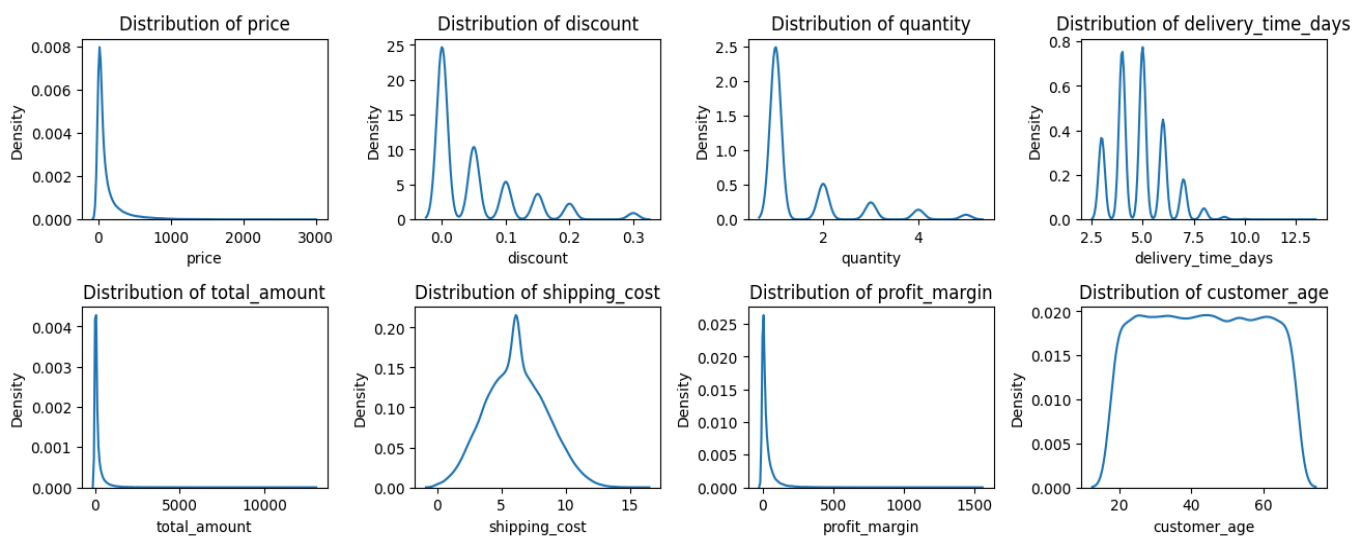
```
for i, col in enumerate(numerical_columns, 1):  
    plt.subplot(4, 4, i) # 4 rows, 4 columns layout  
    sns.kdeplot(x=df[col])  
    plt.title(f'Distribution of {col}')
```

plt.suptitle('KDE Plots of Numerical Columns', fontsize=16, fontweight='bold', y=1.02)

plt.tight\_layout()

plt.show()

## KDE Plots of Numerical Columns



# E-Commerce Sales and Customer Behavior Analysis

```
#Identifying which value to use to treat the outliers in price column  
df['price'].quantile(0.95)
```

```
np.float64(499.31099999999999)
```

```
#Identifying shape of outliers in Price column using boolean indexing & slicing
```

```
pr_threshold = df['price'].quantile(0.95)
```

```
df[df['price'] > pr_threshold].shape
```

```
(1725, 17)
```

```
# Handling the outlier values using 95th quartile using clip()
```

```
df['price'] = df['price'].clip(upper=pr_threshold)
```

```
# Since the price changed, changing the correct values in total_amount
```

```
df['total_amount'] = df['price'] * df['quantity'] * (1 - df['discount'])
```

```
#Stats after handled outliers
```

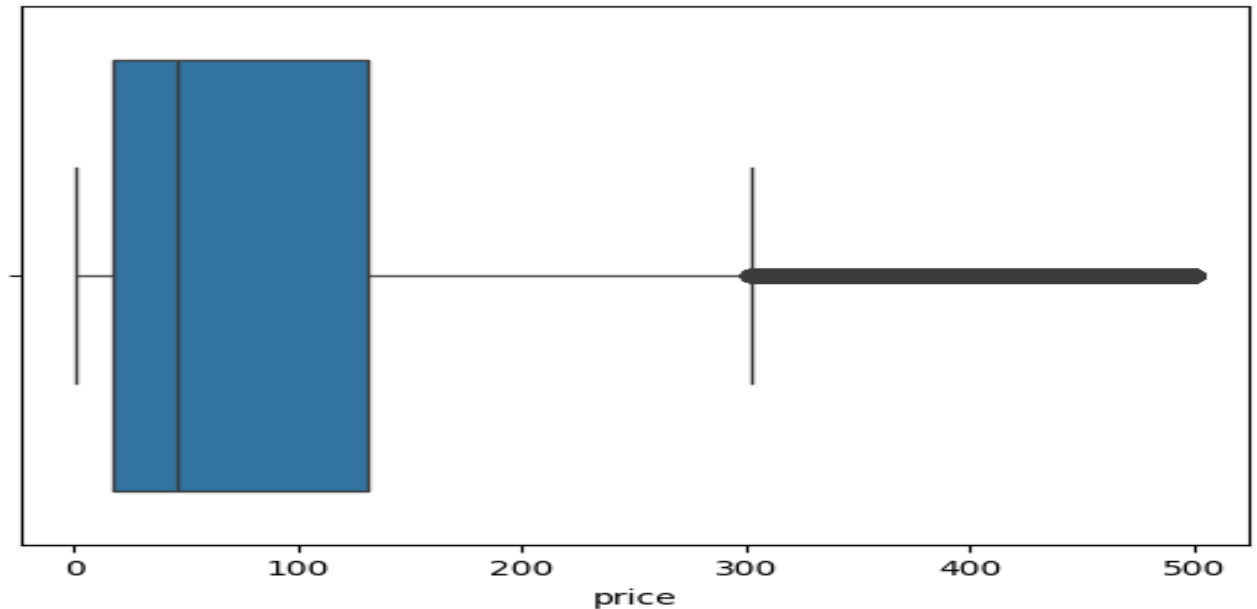
```
df[['price', 'total_amount']].describe()
```

|       | price        | total_amount |
|-------|--------------|--------------|
| count | 34500.000000 | 34500.000000 |
| mean  | 104.541150   | 148.320673   |
| std   | 134.084876   | 243.603391   |
| min   | 1.010000     | 0.816000     |
| 25%   | 16.690000    | 19.711500    |
| 50%   | 45.660000    | 56.851027    |
| 75%   | 130.950000   | 168.649500   |
| max   | 499.311000   | 2496.555000  |

# E-Commerce Sales and Customer Behavior Analysis

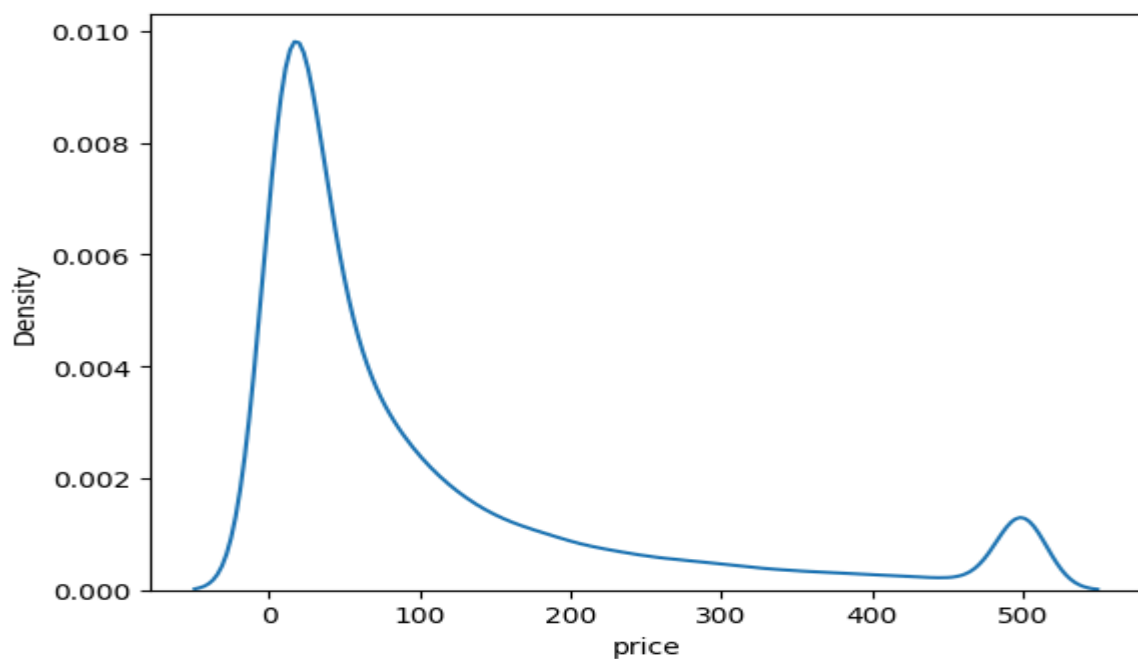
# Boxplot after handling outliers

```
sns.boxplot(x=df['price'])  
plt.show()
```



# KDE plot after handling outliers

```
sns.kdeplot(x=df['price'])  
plt.show()
```



**Note: Some outliers still remain, but this is expected in an e-commerce dataset and does not pose a concern.**

# E-Commerce Sales and Customer Behavior Analysis

# Feature Transformation

# Date type conversion

```
df['order_date'] = pd.to_datetime(df['order_date'])
```

# Adding 3 more columns - Year, Month and day for analysis purposes

```
df['year'] = df['order_date'].dt.year  
df['month'] = df['order_date'].dt.month  
df['day'] = df['order_date'].dt.day
```

```
df.head()
```

| year | month | day |
|------|-------|-----|
| 2023 | 12    | 23  |
| 2025 | 4     | 3   |
| 2024 | 10    | 8   |
| 2024 | 9     | 14  |
| 2024 | 12    | 21  |

## Stage 2 Outcome:

After completing this stage, the dataset becomes clean, consistent, and ready for analysis. Missing values and potential data quality issues are addressed, and useful features are created to support deeper exploratory analysis in the next stage.

# saving the cleaned dataset as excel file

```
df.to_excel('E-commerce_sales_dataset_Cleaned.xlsx', index=False)
```

# E-Commerce Sales and Customer Behavior Analysis

## Stage 3 – EDA and Visualizations

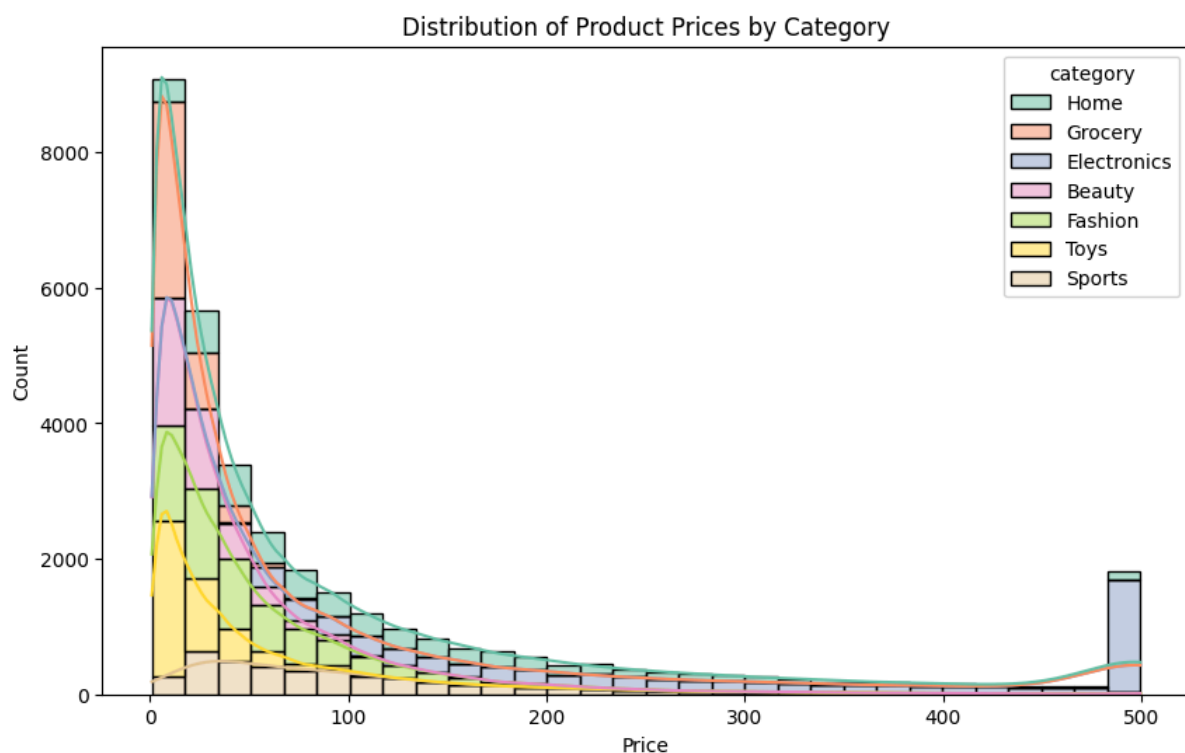
### Univariate Analysis

#### # Chart 1: Price Distribution using Histogram Plot

```
plt.figure(figsize=(10,6))
```

```
sns.histplot(  
    data=df,  
    x='price',  
    bins=30,  
    kde=True,  
    hue='category',    # adds grouping  
    multiple='stack',  # stacks distributions  
    palette='Set2',    # color style  
    stat='count'       # shows frequency  
)
```

```
plt.title("Distribution of Product Prices by Category")  
plt.xlabel("Price")  
plt.ylabel("Count")  
plt.show()
```



# E-Commerce Sales and Customer Behavior Analysis

## Interpretation of above chart

### Chart Overview:

A histogram with KDE is used to visualize the distribution of product prices. The hue parameter is used to differentiate price distributions across product categories.

### Key Observations:

The distribution remains right-skewed, with most products priced lower, while some categories contribute to higher price ranges.

### Features Used:

The price column is used as the main variable, and category is used as a grouping feature to compare how pricing varies across different product categories.

### Business Insight:

Different product categories have varying price ranges, but overall, the majority of products fall into the lower price segment, indicating a focus on affordable items.

# E-Commerce Sales and Customer Behavior Analysis

## # Chart 2: **Category Distribution using Count Plot**

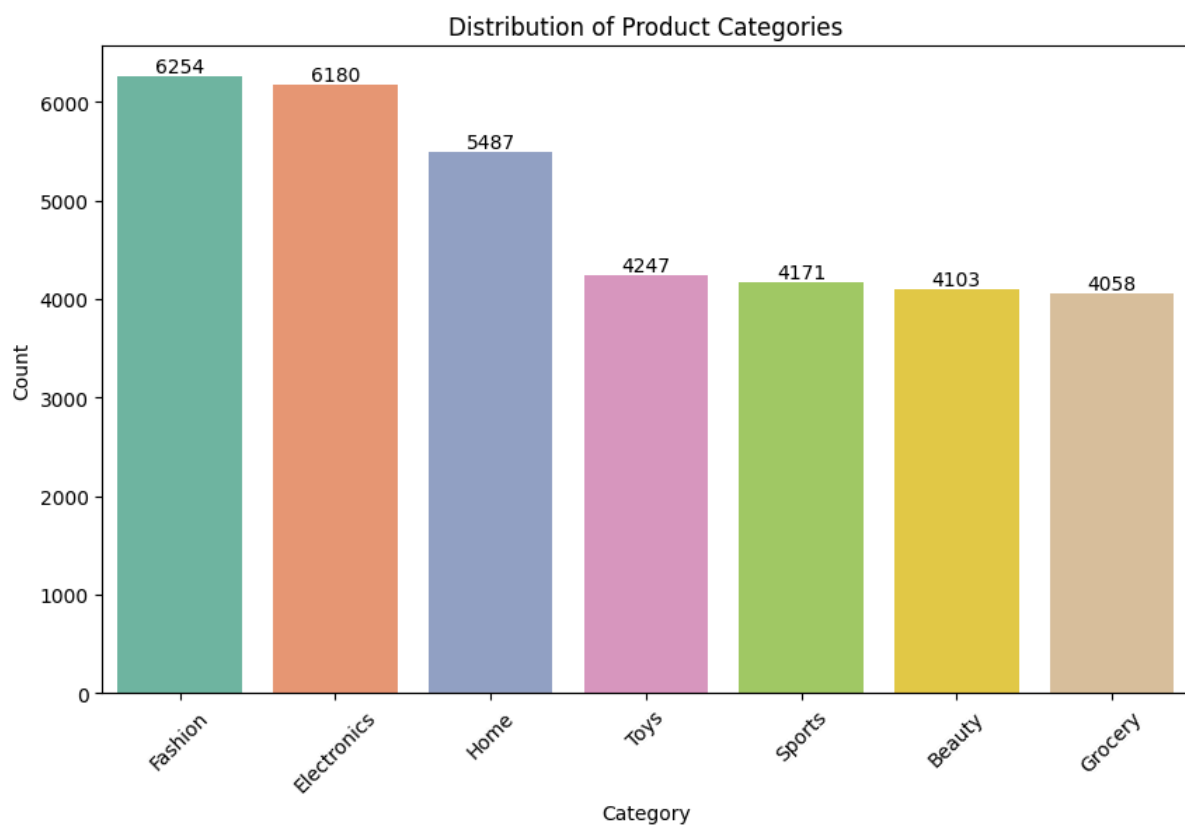
```
plt.figure(figsize=(10,6))

ax = sns.countplot(
    data=df,
    x='category',
    palette='Set2',
    order=df['category'].value_counts().index
)

# Adding bar labels
for container in ax.containers:
    ax.bar_label(container)

plt.title("Distribution of Product Categories")
plt.xlabel("Category")
plt.ylabel("Count")
plt.xticks(rotation=45)

plt.show()
```





# E-Commerce Sales and Customer Behavior Analysis

## Interpretation of above chart

### Chart Overview:

A countplot is used to visualize the frequency of each product category in the dataset. It helps understand how the data is distributed across different categories.

### Key Observations:

Some categories have significantly higher counts compared to others, indicating that certain product types are more frequently purchased or listed in the dataset.

### Features Used:

The category column is used as it represents the classification of products, and it is important for analyzing product popularity and sales distribution.

### Business Insight:

The business appears to focus more on specific product categories, which may indicate higher demand or better sales performance in those segments.

# E-Commerce Sales and Customer Behavior Analysis

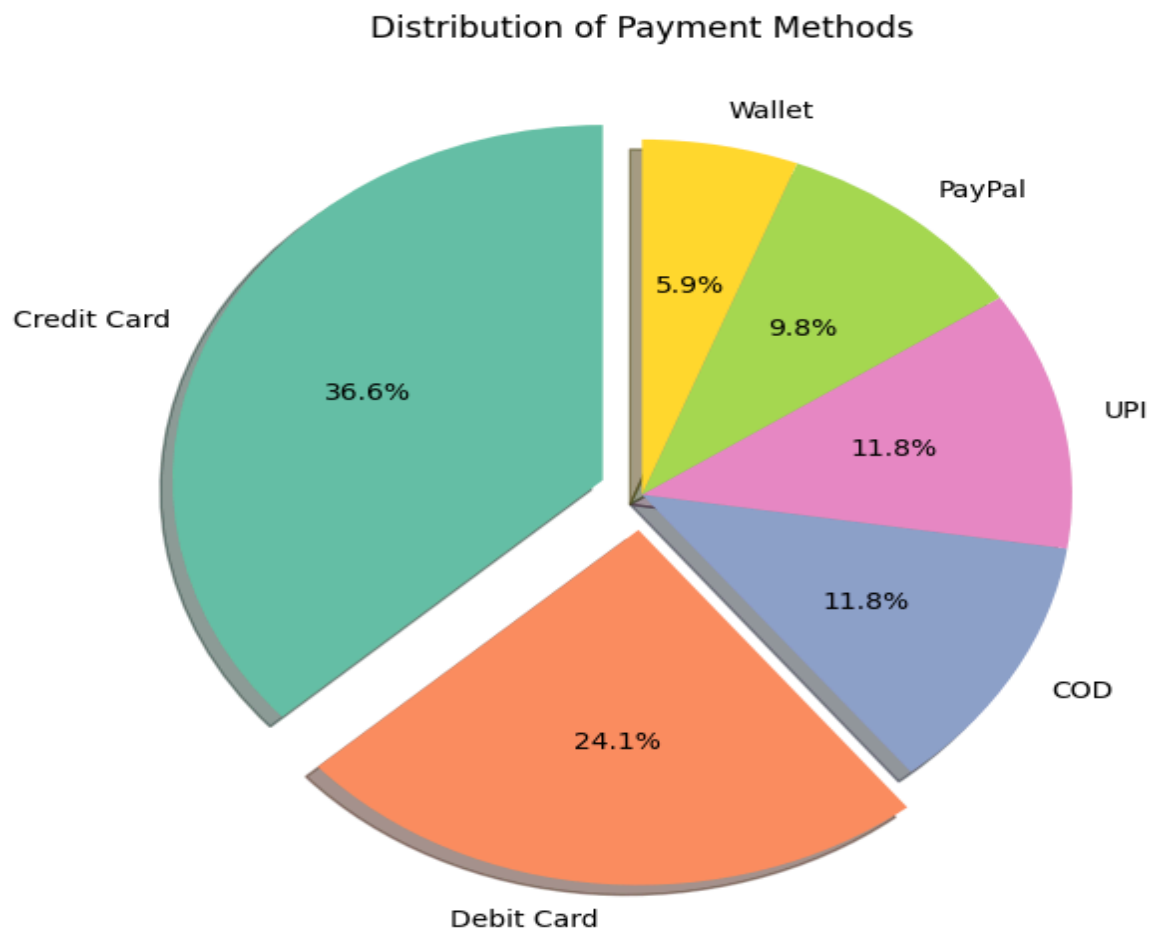
# Chart 3: **Payment Method Distribution using Pie Plot**

```
plt.figure(figsize=(7,7))

payment_counts = df['payment_method'].value_counts()

plt.pie(
    payment_counts,
    labels=payment_counts.index,
    autopct='%1.1f%%',
    colors=sns.color_palette('Set2'),
    startangle=90,
    explode=[0.1,0.1,0,0,0,0],
    shadow=True
)

plt.title("Distribution of Payment Methods")
plt.show()
```



# E-Commerce Sales and Customer Behavior Analysis

## Interpretation of above chart

### Chart Overview:

A pie chart is used to represent the proportion of different payment methods, with the most frequently used method highlighted using the explode feature.

### Key Observations:

The highlighted segment clearly shows the most preferred payment method, while the remaining methods contribute smaller shares.

### Features Used:

The `payment_method` column is used to analyze customer transaction preferences, and the explode feature is applied to emphasize the dominant category.

### Business Insight:

The business can prioritize and optimize the most used payment method to improve customer satisfaction and ensure smoother transaction experiences.

# E-Commerce Sales and Customer Behavior Analysis

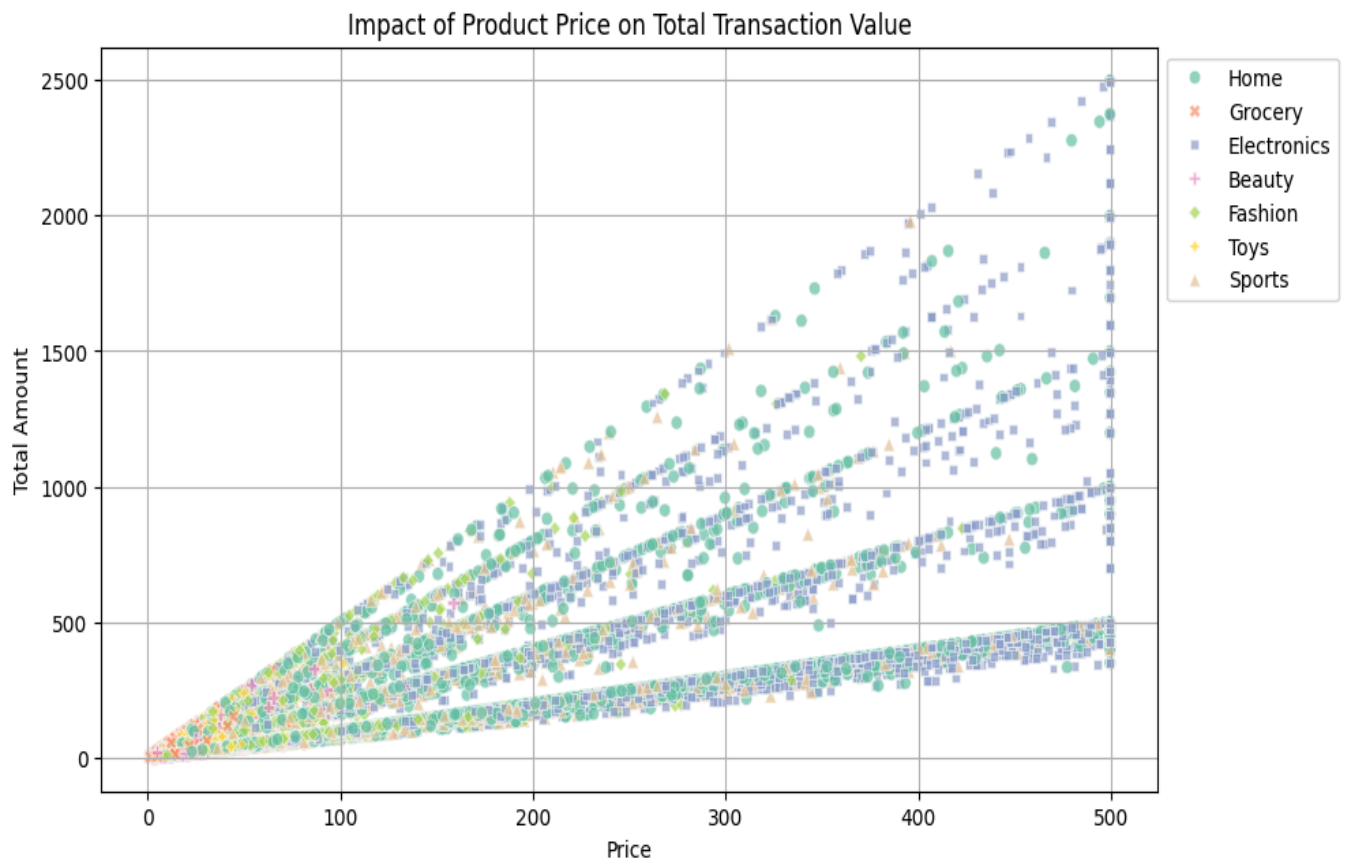
## Bivariate Analysis

Chart 4: Impact of Product Price on Total Transaction Value using scatter plot

```
plt.figure(figsize=(10,6))
```

```
sns.scatterplot(  
    data=df,  
    x='price',  
    y='total_amount',  
    hue='category',  
    palette='Set2',  
    alpha=0.7,  
    style='category',    # adds different markers  
    markers=True        # enables marker variation  
)
```

```
plt.title("Impact of Product Price on Total Transaction Value")  
plt.xlabel("Price")  
plt.ylabel("Total Amount")  
plt.legend(bbox_to_anchor=(1, 1))  
plt.grid(True)  
plt.show()
```



# E-Commerce Sales and Customer Behavior Analysis

## Interpretation of above chart

### Chart Overview:

A scatter plot is used to analyze the relationship between price and total\_amount. It helps visualize how changes in product price impact the total transaction value. The hue parameter is used to differentiate product categories.

### Key Observations:

There is a positive relationship between price and total amount, meaning as the price increases, the total transaction value also tends to increase. However, the spread indicates variability due to quantity and discount.

### Features Used:

The price and total\_amount columns are used as they directly represent product cost and total transaction value. The category feature is added to observe how this relationship varies across different product types.

### Business Insight:

Higher-priced products contribute to larger transaction values, but variations suggest that factors like quantity and discounts also influence total sales. This helps the business understand revenue drivers.

# E-Commerce Sales and Customer Behavior Analysis

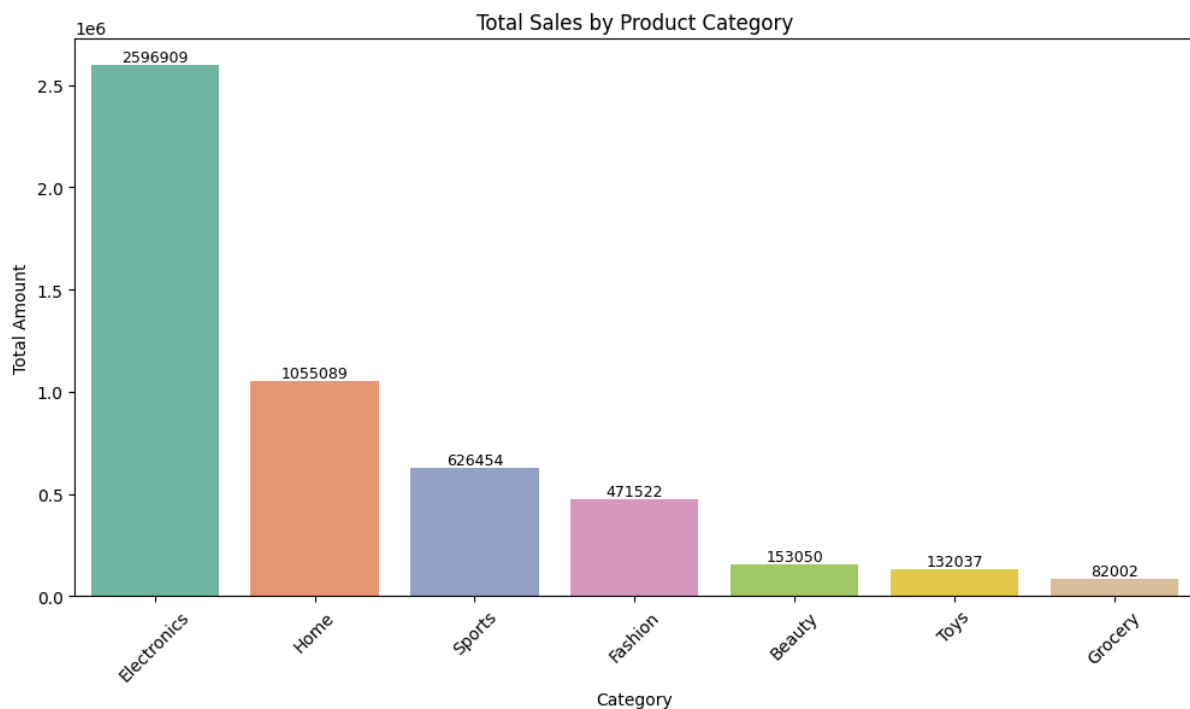
# Chart 5: **Total Sales by Product Category** using Bar plot

```
plt.figure(figsize=(12,6))

ax = sns.barplot(
    data=df,
    x='category',
    y='total_amount',
    estimator=sum,          # total sales per category
    palette='Set2',
    errorbar=None,
    order=df.groupby('category')['total_amount'].sum().sort_values(ascending=False).index #
    # arranging categories based on total sales
)

# Adding bar labels
for container in ax.containers:
    ax.bar_label(container, fmt='%.0f', fontsize=9)

plt.title("Total Sales by Product Category")
plt.xlabel("Category")
plt.ylabel("Total Amount")
plt.xticks(rotation=45)
plt.show()
```



# E-Commerce Sales and Customer Behavior Analysis

## Interpretation of above chart

### Chart Overview:

A bar plot is used to compare the total sales (total\_amount) across different product categories. The estimator=sum is applied to aggregate total revenue for each category.

### Key Observations:

Some categories generate significantly higher total sales compared to others, indicating uneven contribution across product segments.

### Features Used:

The category column is used to group products, while total\_amount is used as it represents the revenue generated from each transaction.

### Business Insight:

The business can identify top-performing categories that drive the majority of revenue and focus on expanding or promoting those segments for increased profitability.

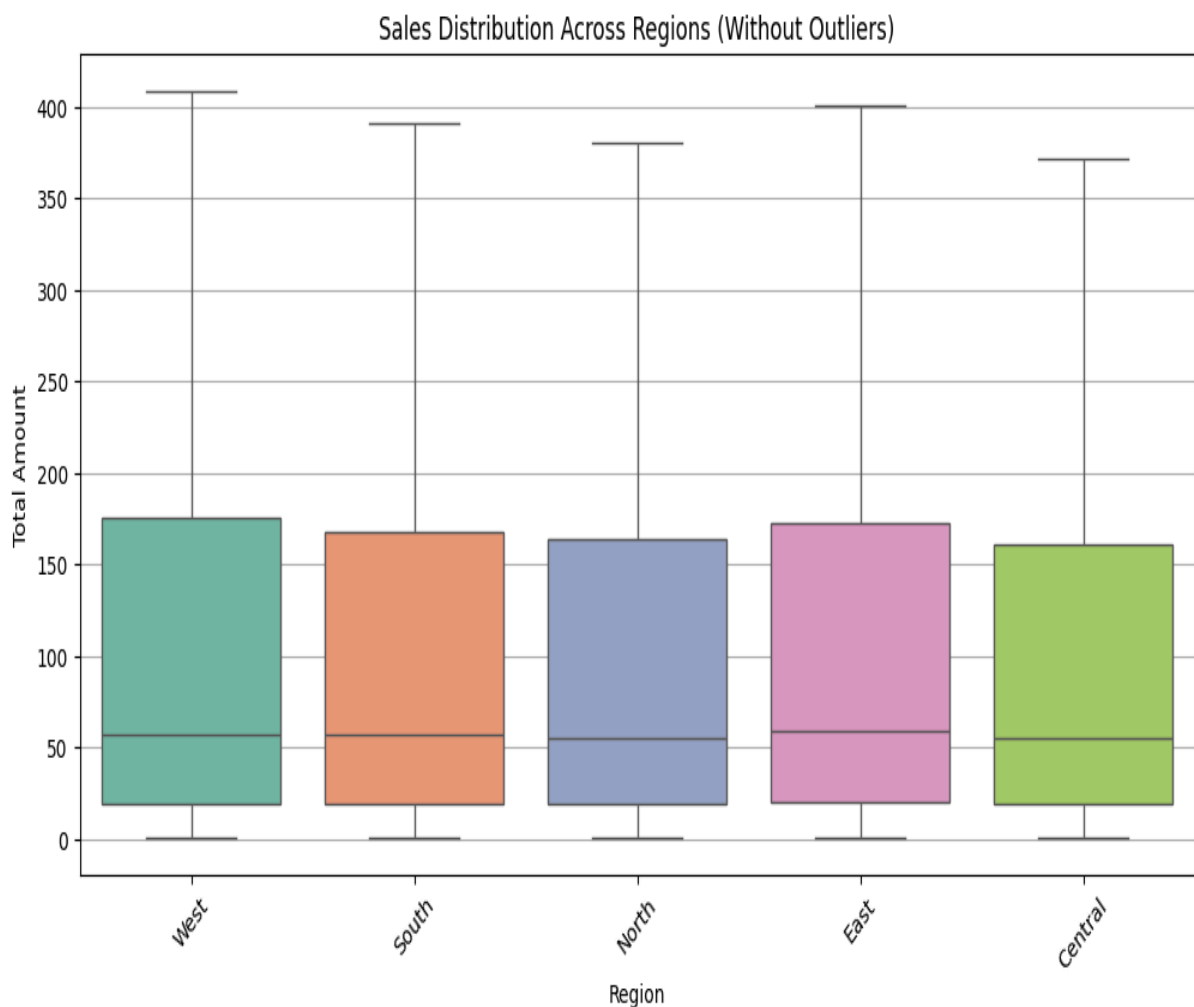
# E-Commerce Sales and Customer Behavior Analysis

# Chart 6: **Sales Distribution Across Regions (Without Outliers)** using Box Plot

```
plt.figure(figsize=(12,6))
```

```
sns.boxplot(  
    data=df,  
    x='region',  
    y='total_amount',  
    palette='Set2',  
    showfliers=False # removes outliers from display  
)
```

```
plt.title("Sales Distribution Across Regions (Without Outliers)")  
plt.xlabel("Region")  
plt.ylabel("Total Amount")  
plt.xticks(rotation=45)  
plt.grid(axis='y')  
plt.show()
```





# E-Commerce Sales and Customer Behavior Analysis

## Interpretation of above chart

### Chart Overview:

A box plot is used to visualize the distribution of total sales across regions, with outliers removed to focus on the central data distribution.

### Key Observations:

The spread of values across regions becomes clearer without extreme values, making it easier to compare median sales and variability.

### Features Used:

The region column is used for comparison across geographical areas, while `total_amount` represents transaction values.

### Business Insight:

By removing outliers from visualization, the business can better understand typical sales patterns across regions and make more consistent comparisons.

# E-Commerce Sales and Customer Behavior Analysis

Chart 7A: **Monthly Sales Trend**

```
plt.figure(figsize=(12,6))
```

```
# Extract month
```

```
df['month'] = df['order_date'].dt.month
```

```
# Aggregate sales by month
```

```
monthly_sales = df.groupby('month')['total_amount'].sum()
```

```
# Plot
```

```
plt.plot(  
    monthly_sales.index,  
    monthly_sales.values,  
    marker='o'  
)
```

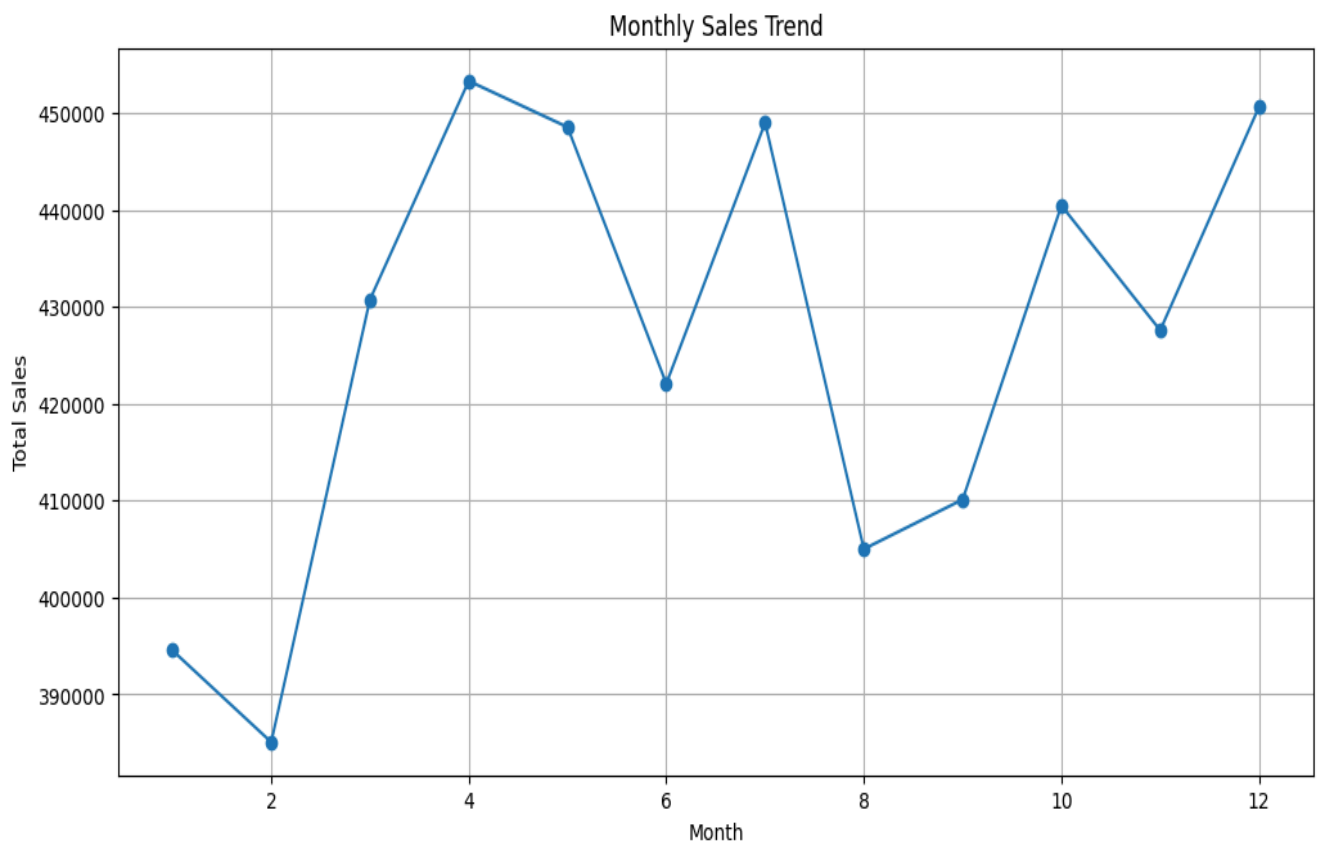
```
plt.title("Monthly Sales Trend")
```

```
plt.xlabel("Month")
```

```
plt.ylabel("Total Sales")
```

```
plt.grid(True)
```

```
plt.show()
```



# E-Commerce Sales and Customer Behavior Analysis

## # Chart 7B: Yearly Sales Trend

```
plt.figure(figsize=(10,5))

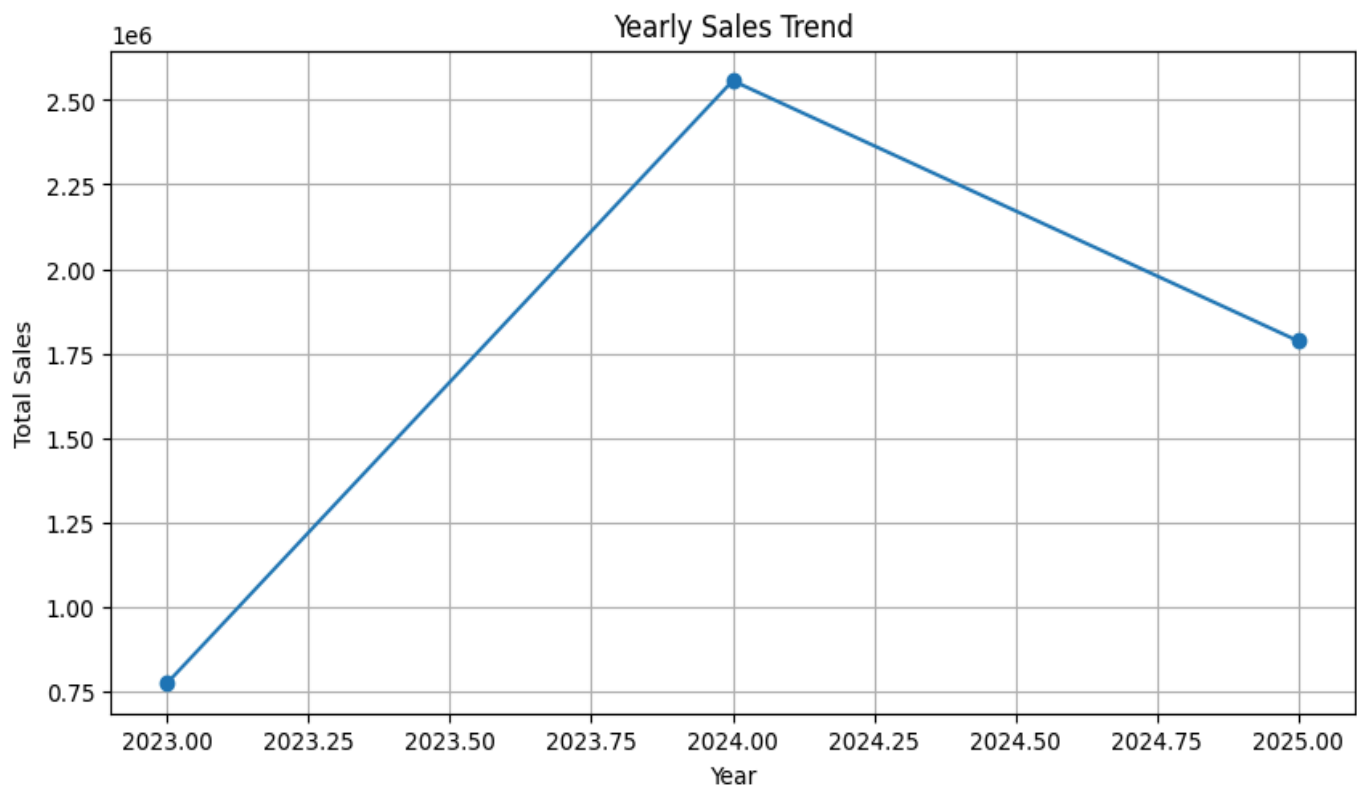
# Extract year
df['year'] = df['order_date'].dt.year

# Aggregate sales by year
yearly_sales = df.groupby('year')['total_amount'].sum()

# Plot
plt.plot(
    yearly_sales.index,
    yearly_sales.values,
    marker='o'
)

plt.title("Yearly Sales Trend")
plt.xlabel("Year")
plt.ylabel("Total Sales")

plt.grid(True)
plt.show()
```



## E-Commerce Sales and Customer Behavior Analysis

### Interpretation of above charts (Monthly + Yearly Sales Trend)

#### Chart Overview:

Line charts are used to analyze sales trends over time at both monthly and yearly levels. Monthly analysis captures short-term variations, while yearly analysis provides an overall long-term trend.

#### Key Observations:

The monthly trend shows fluctuations in sales, indicating periods of higher and lower customer activity, which may reflect seasonal demand patterns. The yearly trend highlights the overall direction of sales performance, showing whether the business is growing, stable, or declining over time.

#### Features Used:

The `order_date` column is used to extract both month and year, while `total_amount` is used as it represents total sales aggregated over time.

#### Business Insight:

The combined analysis helps the business identify seasonal trends as well as long-term growth patterns. This enables better planning for inventory, marketing campaigns, and resource allocation during peak and low sales periods.

# E-Commerce Sales and Customer Behavior Analysis

## Multivariate Analysis

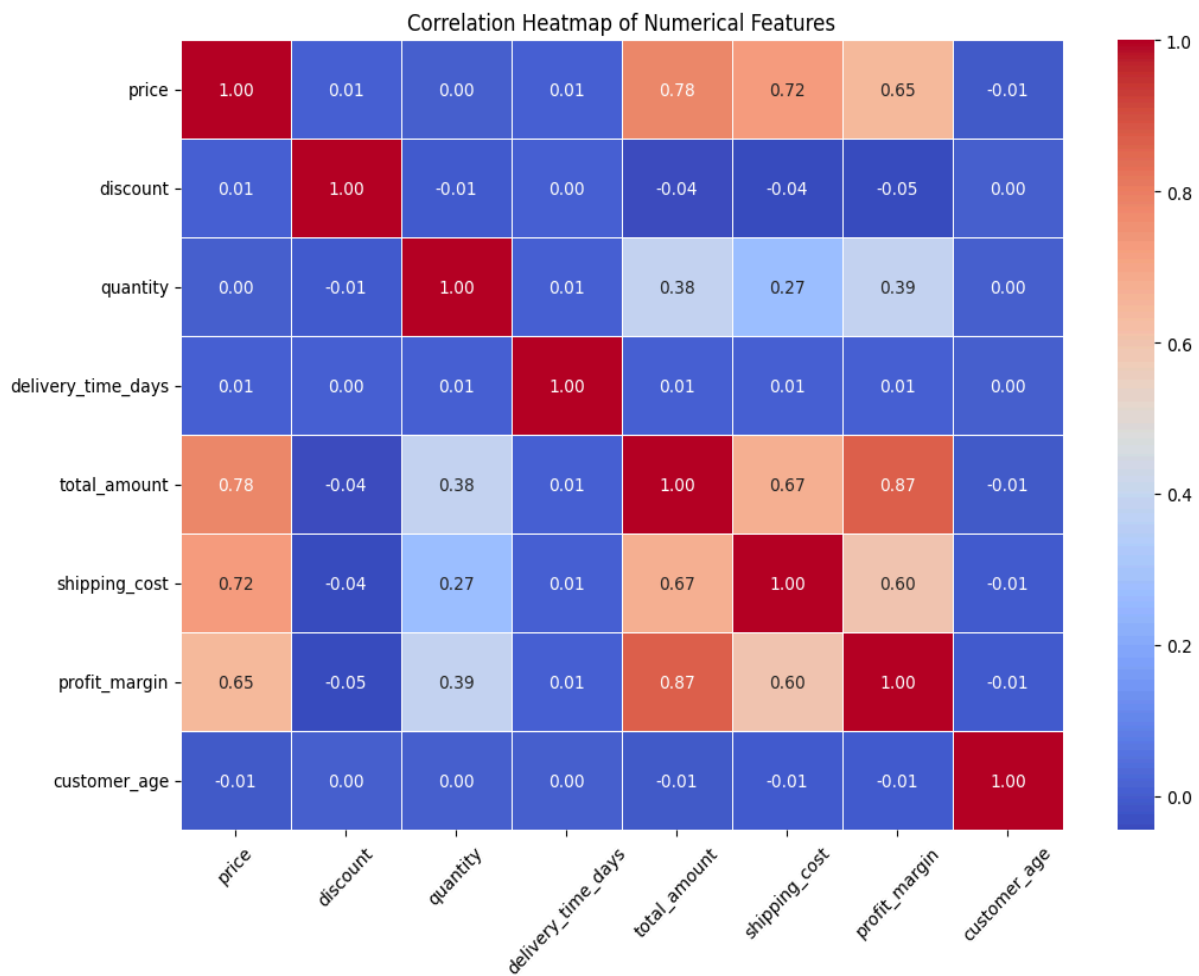
### # Chart 8: Correlation Heatmap of Numerical Features

```
plt.figure(figsize=(12,8))

# Select numerical columns
corr = df[numerical_columns].corr()

sns.heatmap(
    corr,
    annot=True,      # shows values
    cmap='coolwarm', # color theme
    fmt='.2f',       # 2 decimal points
    linewidths=0.5
)

plt.title("Correlation Heatmap of Numerical Features")
plt.xticks(rotation=45)
plt.show()
```



# E-Commerce Sales and Customer Behavior Analysis

## Interpretation of above chart

### Chart Overview:

A heatmap is used to visualize the correlation between multiple numerical variables in the dataset. It helps identify the strength and direction of relationships between features.

### Key Observations:

Some variables show strong positive correlations, such as between price and total amount, indicating that as one increases, the other also increases. Other variables may show weak or no correlation, suggesting little or no linear relationship.

### Features Used:

All numerical columns such as price, quantity, total\_amount, discount, profit\_margin, and others are used to analyze relationships across multiple variables simultaneously.

### Business Insight:

Understanding correlations helps the business identify key drivers of revenue and performance. Strong relationships between variables can guide pricing strategies, discount planning, and sales optimization.

# E-Commerce Sales and Customer Behavior Analysis

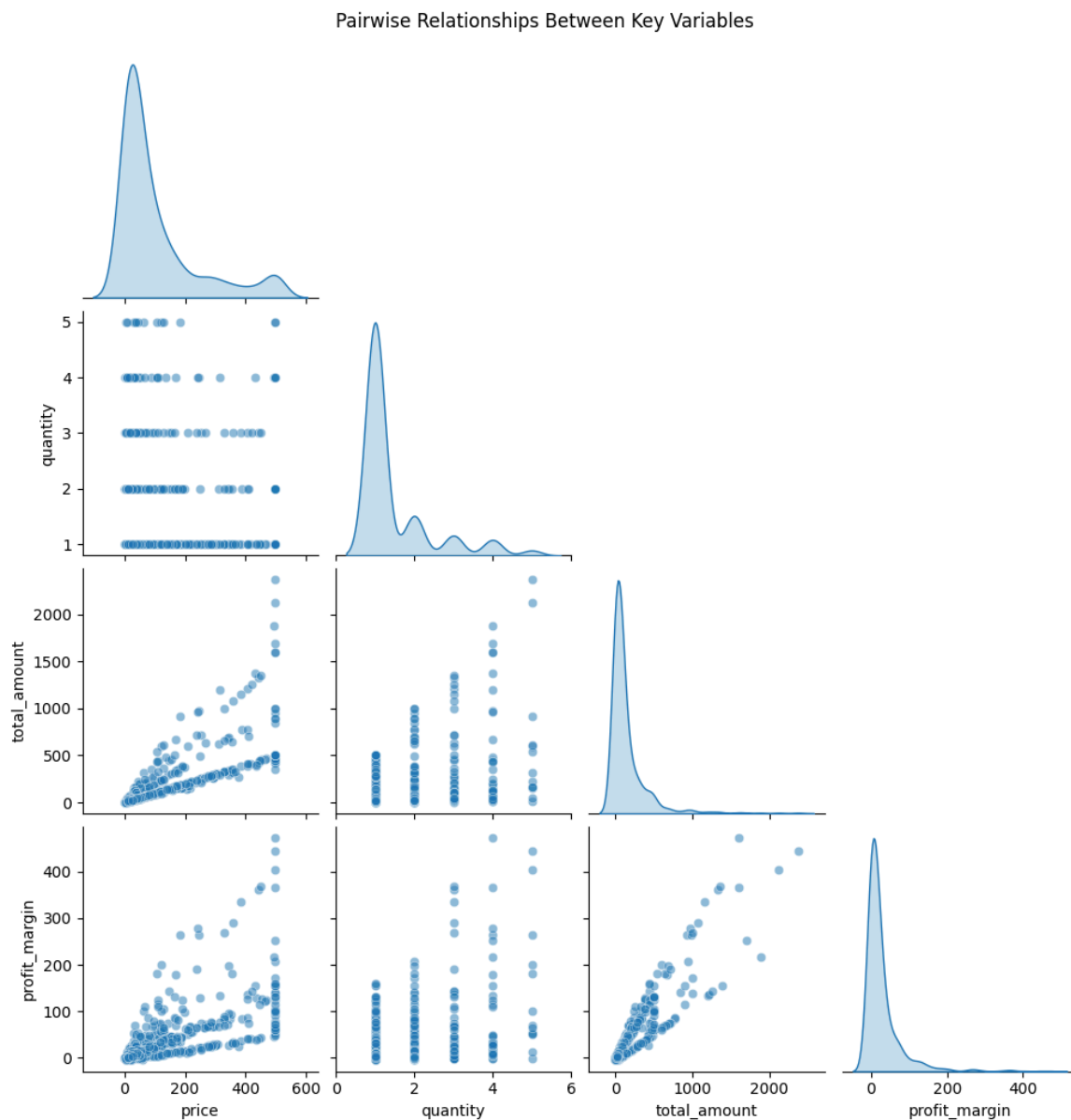
## # Chart 9: Pairwise Relationships Between Key Variables using Pair plot

```
# Selecting only important features (avoid clutter)
selected_cols = ['price', 'quantity', 'total_amount', 'profit_margin']

# Taking a smaller sample for clarity
sample_df = df[selected_cols].sample(800)

sns.pairplot(
    sample_df,
    diag_kind='kde',
    corner=True,      # removes duplicate plots
    plot_kws={'alpha': 0.5}, # reduces overlap
)

plt.suptitle("Pairwise Relationships Between Key Variables", y=1.02)
plt.show()
```



# E-Commerce Sales and Customer Behavior Analysis

## Interpretation of above chart

### Chart Overview:

A pairplot is used to visualize relationships between selected key numerical variables, providing a cleaner and focused view of interactions.

### Key Observations:

Clear relationships can be observed between variables such as price and total amount, while other variables show weaker or more dispersed relationships.

### Features Used:

Key numerical features like price, quantity, total\_amount, and profit\_margin are selected to focus on the most relevant business variables.

### Business Insight:

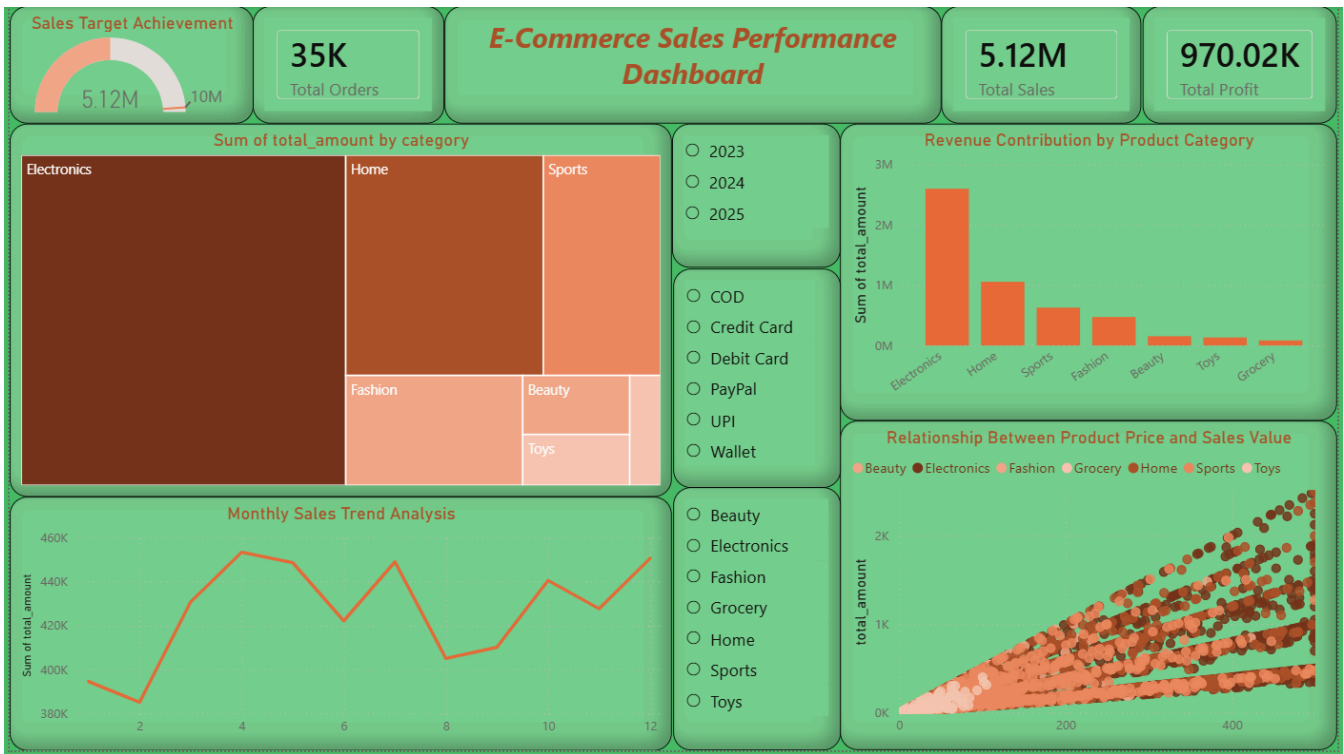
This focused analysis helps identify key drivers of sales and profitability while avoiding unnecessary complexity from less relevant variables.



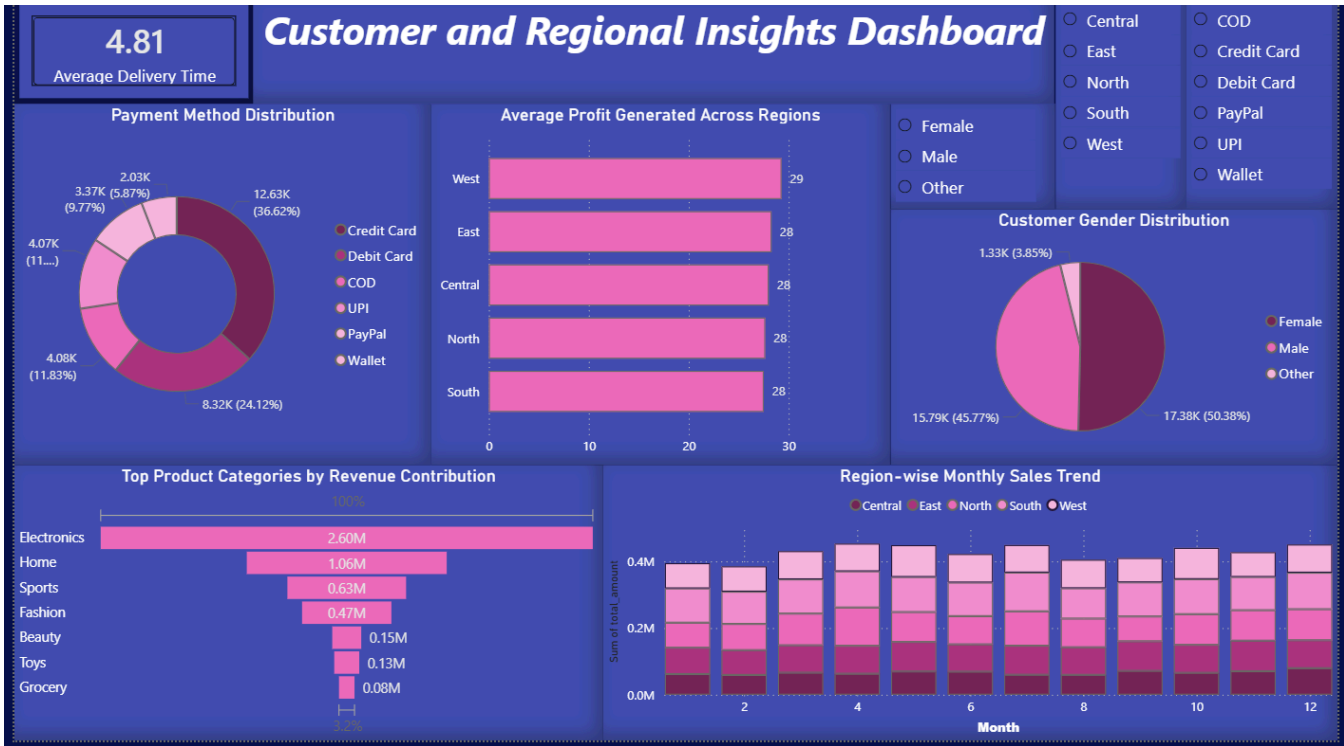
# E-Commerce Sales and Customer Behavior Analysis

## Stage 4 – Documentation, Insights and Presentation

### Dashboard 1: E-Commerce Sales Performance Dashboard



### Dashboard 2: Customer and Regional Insights Dashboard



# E-Commerce Sales and Customer Behavior Analysis

## Dashboards Summary:

The Power BI dashboards provide a comprehensive analysis of e-commerce sales performance, customer behavior, and regional trends. The dashboards combine sales metrics, product performance, customer segmentation, payment preferences, and regional insights into an interactive business intelligence solution. Through visual storytelling and KPI tracking, the dashboards help identify high-performing categories, customer purchasing patterns, and areas for business improvement.

## Key Insights from the Dashboard:

### 1. Electronics Category Dominates Revenue

The Electronics category contributes the highest share of total revenue compared to all other product categories. This indicates strong customer demand and higher transaction values within electronics products.

### 2. Sales Show Consistent Monthly Performance

The monthly sales trend remains relatively stable throughout the year with slight fluctuations during certain months. This suggests steady customer purchasing activity and consistent sales performance.

### 3. Higher Product Prices Lead to Higher Sales Value

The scatter plot reveals a positive relationship between product price and total sales amount. As product prices increase, the transaction value also increases significantly.

### 4. Credit Card is the Most Preferred Payment Method

Among all payment methods, credit card transactions account for the largest proportion of purchases, indicating strong customer preference for digital payment options.

### 5. Regional Profit Performance is Balanced

Average profit generated across regions remains relatively consistent, showing that business performance is distributed evenly across different geographical areas.

## Anomalies / Observations

- \* Electronics products contribute significantly more revenue than categories such as Grocery and Toys.
- \* A few high-value transactions create larger sales spikes in the scatter plot.
- \* Certain months show slightly lower sales trends, indicating possible seasonal demand variations.

# E-Commerce Sales and Customer Behavior Analysis

## Business Recommendations:

### 1. Focus More on Electronics Products

Since Electronics generates the highest revenue, the business can increase marketing campaigns, stock availability, and promotional strategies for this category.

### 2. Strengthen Digital Payment Offers

As credit card payments dominate customer preferences, introducing cashback offers and online payment discounts may improve customer engagement further.

### 3. Improve Low-Performing Categories

Categories such as Grocery and Toys contribute lower revenue. The business can introduce targeted promotions and pricing strategies to improve sales in these segments.

### 4. Use Seasonal Trends for Planning

Monthly sales trends can help businesses prepare inventory and marketing campaigns during peak sales periods to maximize revenue opportunities.

### 6. Optimize Regional Marketing Strategies

Since regional performance is relatively balanced, region-specific promotional campaigns can further strengthen customer reach and profitability.

## Final Story of the Dashboard Visuals:

The dashboard visuals highlight the overall performance of the e-commerce business through sales, profit, customer behavior, and regional analysis. KPI visuals indicate strong revenue generation, profit growth, and a high number of completed orders.

The category-based charts show that Electronics is the top-performing category contributing the highest revenue, while the sales trend visuals indicate consistent customer demand throughout the year. The scatter plot reveals a positive relationship between product price and sales value, showing that higher-priced products generate larger transaction amounts.

Customer and regional visuals show that digital payment methods are highly preferred and that business performance remains balanced across different regions. Overall, the dashboards provide a clear understanding of key sales patterns, customer preferences, and business performance to support data-driven decision-making.